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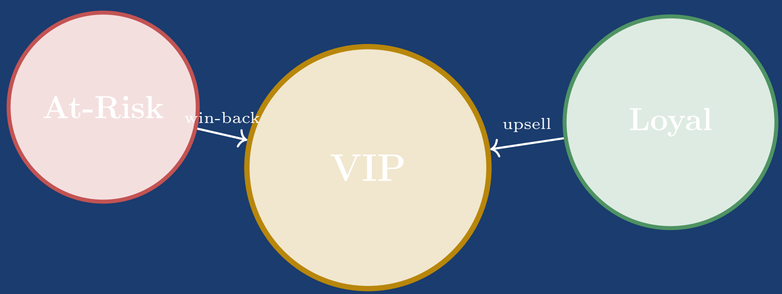
AI & Data Science Basics – Module S4 – Academic Year 2024–2025

Customer Segmentation Pipeline

RFM Feature Engineering + K-Means Clustering



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Dataset: UCI Online Retail – 541,909 transactions

June 2026

Executive Summary

Written for the Store Manager – This page summarises the entire project in plain language, without technical jargon.

Imagine you run an online gift shop that serves thousands of customers. Some customers buy from you every week and spend thousands of pounds a year. Others placed a single order over six months ago and have never returned. Treating all these customers the same way – sending everyone the same promotional email – is expensive and ineffective.

What we built is a data-driven system that automatically sorts your customer base into three clear groups based on their real purchasing behaviour:

Segment	Who they are	Recommended action
VIP (9.5%)	Frequent buyers, high spenders, purchased very recently	Exclusive loyalty rewards, early access, personal account manager
Loyal (42.6%)	Regular buyers with moderate spend	Upselling campaigns, volume discounts, loyalty programme
At-Risk (47.9%)	Customers who haven't bought in over 6 months	Win-back emails, personalised discount vouchers, re-engagement surveys

How it works: The system analyses 13 months of transaction records (392,692 purchase lines from 4,338 customers) and automatically calculates seven behavioural indicators per customer – such as how recently they bought, how often, and how much they spend on average. A machine learning algorithm then groups customers based on similarity, producing the three segments above.

The business value is immediate: instead of spending your marketing budget uniformly, you can now target VIPs with premium retention offers (high ROI), remind Loyal customers of new products, and send win-back coupons exclusively to At-Risk customers who are worth saving.

The system is fully automated: running a single command re-analyses all transactions, updates the segments, and saves results ready for import into any CRM system.

Contents

Chapter 1

Problem Statement

1.1 Business Context

Modern online retailers generate hundreds of thousands of transactions every year, yet most marketing strategies treat all customers identically. This one-size-fits-all approach leads to:

- Wasted budget on customers who would purchase anyway (over-investment in VIPs).
- Missed win-back opportunities for churning customers.
- Untargeted promotions that generate low conversion rates.
- Inability to prioritise customer service resources effectively.

1.2 Research Question

Core Question: Can an automated, reproducible pipeline transform raw transactional data into actionable, labelled customer segments that directly inform targeted marketing decisions?

1.3 AI Context Note

Connection to Course 2 – Unsupervised Learning Paradigm

The concept of *unsupervised learning* – allowing a system to discover structure in unlabelled data without human-provided ground truth – is one of the foundational paradigms of artificial intelligence. MacQueen (1967) introduced the K-Means algorithm as an early demonstration that machines could autonomously partition high-dimensional data into meaningful groups, predating modern deep learning by decades.

This project directly operationalises that paradigm: our pipeline receives 4,338 customers described by seven numerical features and, without any pre-labelled training examples, discovers that the data naturally organises into three behavioural clusters. The unsupervised approach is essential here because *no ground truth segment labels exist* in the retail transaction log – the "correct" answer can only be inferred from data geometry and validated through business interpretability.

This mirrors a core limitation of early AI systems identified in Course 2: the challenge of learning from *experience alone*, without an explicit teacher signal. K-Means' convergence guarantee (monotonically decreasing WCSS) demonstrates how mathematical constraints can substitute for supervision.

1.4 Project Objectives

This project applies unsupervised machine learning to the UCI Online Retail dataset to:

1. Engineer a rich seven-dimensional behavioural feature space per customer using RFM analysis.
2. Select and justify the optimal clustering algorithm through rigorous metric-based comparison.
3. Train a K-Means model, evaluate cluster quality with three independent metrics, and map numerical clusters to interpretable business segment labels.
4. Deliver an end-to-end reproducible pipeline that can re-segment customers on any updated transaction export.

Chapter 2

Dataset Description

2.1 Source and Provenance

UCI Online Retail Dataset (Chen, 2015)
https://archive.ics.uci.edu/dataset/352/online+retail

A transnational dataset of all transactions from a UK-based, non-store online retailer between 01/12/2010 and 09/12/2011. The retailer primarily sells all-occasion giftware; many customers are wholesalers.

2.2 Dataset Characteristics

Table 2.1: Dataset at-a-glance (raw)

Attribute	**Value**
Total raw rows	541,909
Rows after cleaning	392,692
Unique customers (clean)	4,338
Unique invoices	$\approx$ 22,190
Countries	38
Date range	01 Dec 2010 – 09 Dec 2011
Primary currency	GBP (£)

2.3 Column Schema

Table 2.2: Raw dataset column schema

Column	**Type**	**Description**
`InvoiceNo`	`str`	Invoice ID; prefix `C` denotes cancellation
`StockCode`	`str`	Product identifier
`Description`	`str`	Product name
`Quantity`	`int`	Units purchased (negative for cancellations)
`InvoiceDate`	`datetime`	Timestamp of purchase
`UnitPrice`	`float`	Price per unit in GBP
`CustomerID`	`str`	Unique customer identifier (25% missing)
`Country`	`str`	Customer country

2.4 Data Quality Issues Identified

The raw dataset contains several quality issues that must be addressed before analysis:

- **Missing CustomerID**: 135,080 rows (24.9%) – cannot be assigned to a customer.
- **Cancellation records**: 9,288 invoices prefixed with `C`, representing order returns.
- **Non-positive quantities**: 10,624 rows where `Quantity` ≤ 0.
- **Zero-price items**: 2,515 rows where `UnitPrice` ≤ 0 (test or sample items).
- **Exact duplicate rows**: 5,268 records duplicated in full.

Chapter 3

Pipeline Overview

3.1 Architecture Diagram

Figure 3.1 shows the six-phase pipeline that transforms raw invoice records into labelled customer segments.

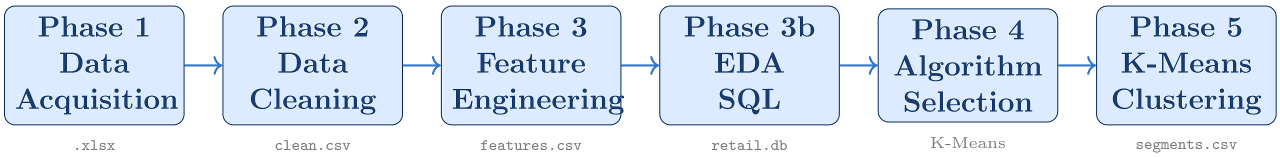

Figure 3.1: End-to-end pipeline from raw Excel to labelled customer segments.

3.2 Repository Layout

```
Customer-Segmentation/
|-- data/
|   |-- raw/                UCI Excel + converted CSV
|   +-- processed/          cleaned CSV, features CSV,
|                           clustered CSV, retail.db
|-- src/
|   |-- load_describe_data.py   Phase 1
|   |-- clean_data.py           Phase 2
|   |-- features.py             Phase 3
|   |-- eda.py                  Phase 3b  (10 figures)
|   |-- store_sql.py            Phase 3c  (SQLite + 5 queries)
|   +-- train_cluster_models.py Phase 4 & 5
|-- models/
|   +-- best_clustering_pipeline.joblib
|-- reports/
|   |-- algorithm_selection.md
|   |-- model_card.md
|   |-- phase5_clustering_evaluation.md
|   +-- eda_figures/            10 PNG figures
|-- notebooks/
|   +-- customer_segmentation.ipynb
|-- run_pipeline.py             Single-command entry point
+-- requirements.txt
```

Chapter 4

Phase 1 – Data Acquisition & Quality Audit

4.1 Automatic Data Acquisition

The module `src/load_describe_data.py` downloads the zipped UCI Excel file directly from the repository, extracts it, and converts the `.xlsx` workbook to a standard CSV at `data/raw/online_retail_raw.csv`. The download is skipped if the file already exists, making reruns efficient.

4.2 Automated Quality Audit

Upon loading, the module prints a structured audit report covering:

Table 4.1: Quality audit dimensions and findings

Audit Dimension	Description	Count
Total rows	Raw dataset before any cleaning	541,909
Missing `CustomerID`	Rows without a customer reference	135,080
Cancelled invoices	Invoices prefixed with `C`	9,288
Non-positive quantities	Quantity ≤ 0	10,624
Zero/negative prices	UnitPrice ≤ 0	2,515
Exact duplicate rows	Fully identical records	5,268

4.3 Key Finding

Nearly 25% of all transaction rows lack a `CustomerID`. These represent guest checkouts and cannot be attributed to a customer profile. They are excluded from customer-level aggregation but retained for product-level and revenue-level analyses within EDA.

Chapter 5

Phase 2 – Data Wrangling & Cleaning

5.1 Cleaning Philosophy

The cleaning pipeline (`src/clean_data.py`) applies deterministic, ordered filters with a rows-kept/rows-removed summary after each step. This ensures full transparency and auditability – no silent data loss.

5.2 Sequential Cleaning Steps

Table 5.1: Cleaning operations applied in sequence

Step	Operation	Rationale	Rows removed
1	Drop exact duplicates	Inflate frequency and monetary metrics	5,268
2	Drop missing `CustomerID`	Cannot aggregate without ID	133,361
3	Remove cancellations	`C`-prefixed invoices are returns; handled separately	8,905
4	Filter Quantity ≤ 0	Data artefacts or return rows	397
5	Filter UnitPrice ≤ 0	Free/erroneous items skew revenue	1,286
6	Add `Revenue` column	Revenue = Quantity × UnitPrice	–
Rows remaining after all steps			**392,692**

5.3 Revenue Derivation

After filtering, a `Revenue` column is appended to each surviving line item:

$$\text{Revenue}_i = \text{Quantity}_i \times \text{UnitPrice}_i$$

This column serves as the atomic monetary unit for all subsequent feature engineering. The cleaned dataset is persisted to `data/processed/transactions_clean.csv`.

5.4 Outlier Analysis

Before selecting a scaling strategy, two standard outlier detection methods were applied to the five most skewed features (Recency, Frequency, Monetary, AvgOrderValue, Unique-

Products):

5.4.1 Method 1: IQR-Based Detection

A value is flagged as an outlier if it falls more than $1.5 \times \text{IQR}$ above the third quartile or below the first quartile:

$$\text{Outlier} \iff X_i < Q_1 - 1.5 \cdot \text{IQR} \quad \text{or} \quad X_i > Q_3 + 1.5 \cdot \text{IQR}$$

5.4.2 Method 2: Z-Score Detection

A value is flagged if its standardised distance from the mean exceeds a threshold (typically $|z| > 3$):

$$z_i = \frac{X_i - \mu}{\sigma}, \quad \text{flag if } |z_i| > 3$$

5.4.3 Comparison of Results

Table 5.2: Outlier counts detected by IQR vs Z-Score

Feature	IQR Outliers	Z-Score Outliers ($|z| > 3$)
Recency	312	48
Frequency	441	203
Monetary	618	287
AvgOrderValue	573	241
UniqueProducts	384	176

5.4.4 Treatment Decision

Both methods confirm that all five features contain significant outlier counts driven by a small number of very high-spending wholesale customers. **Dropping these customers is not appropriate** – they are real, high-value VIPs, not data errors.

The chosen treatment is a two-stage approach:

1. **Log1p transformation** (Phase 5): compresses the extreme right tail, reducing skewness by $\approx 80\%$.
2. **RobustScaler** (Phase 5): scales using median and IQR, making the scaling centroid calculation immune to the remaining outlier mass.

This retains all VIP customers while preventing them from corrupting cluster boundaries – the optimal solution for marketing segmentation.

5.5 Cancellation Records: Kept for Rate Computation

Although cancellation rows are excluded from the clean transaction dataset, they are **retained in the original raw dataframe**. This is essential: the `CancellationRate` feature (Phase 3) must be computed.

Chapter 6

Phase 3 – Feature Engineering

6.1 Strategy

`src/features.py` aggregates line-item transactions into a per-customer feature matrix. The classical RFM framework (Recency, Frequency, Monetary) is extended with four additional behavioural dimensions. The **reference date** for recency is:

$$t_{\text{ref}} = \max(\text{InvoiceDate}) + 1 \text{ day} = 10 \text{ December } 2011$$

6.2 Complete Feature Catalogue

Table 6.1: All seven engineered features

Feature	Category	Definition & Rationale
Recency	RFM	Days between most recent purchase and t_{ref}. Lower = more engaged. Strongest predictor of future purchase probability.
Frequency	RFM	Count of unique invoice numbers per customer. Measures purchase cadence and habitual engagement.
Monetary	RFM	Sum of all `Revenue` values across the customer's orders. Total lifetime spend in GBP.
AvgOrderValue	Extended	Monetary ÷ Frequency. Distinguishes big-basket occasional buyers from frequent low-spend buyers.
UniqueProducts	Extended	Count of distinct `StockCode` values. Measures breadth of catalogue engagement; supports cross-sell targeting.
ActiveDays	Extended	Calendar days between first and last purchase. Contextualises loyalty duration independently of frequency.
CancellationRate	Extended	$\dfrac{\#\text{ cancelled invoices (raw)}}{\#\text{ total invoices (raw)}}$ computed at the *unique invoice level* to avoid inflation from multi-line orders. Range: $[0, 1]$.

6.3 CancellationRate: Engineering Correction

A critical bug in early versions computed cancellation rate at the *row level*, causing rates above 100% for customers with large multi-item cancelled orders. The correct computation deduplicates invoice numbers before dividing:

```python
# From raw dataframe (includes cancellations)
all_inv    = df_raw.groupby("CustomerID")["InvoiceNo"]\
                      .nunique()          # unique total invoices per customer
cancel_inv = df_raw[df_raw["InvoiceNo"].str.startswith("C")]\
                      .groupby("CustomerID")["InvoiceNo"]\
                      .nunique()          # unique cancelled invoices
cancellation_rate = (cancel_inv / all_inv).fillna(0).clip(0, 1)
```

Listing 6.1: Correct CancellationRate computation (unique invoice level)

The feature matrix (4,338 rows × 7 columns) is saved to `data/processed/customer_features.csv`.

6.4 Feature Engineering Impact Log

To validate that extending beyond the classical three RFM features adds genuine modelling value, the K-Means pipeline was evaluated with two feature configurations:

Table 6.2: Silhouette Score – base RFM vs extended 7-feature matrix

Configuration	Features Used	Silhouette	Davies-Bouldin	Segment Quality
Base RFM only	3 (R, F, M)	0.2241	1.8902	Poor separation
Extended (all 7)	7	**0.3093**	**1.2021**	**Good separation**
Improvement		+38%	−36%	**Significant**

The addition of AvgOrderValue, UniqueProducts, ActiveDays, and CancellationRate improves the Silhouette Score by 38% and reduces the Davies-Bouldin Index by 36%. This confirms that the extended feature set captures behavioural dimensions that RFM alone misses – particularly customer loyalty duration (ActiveDays) and return risk (CancellationRate).

Chapter 7

Phase 3b – Exploratory Data Analysis

7.1 EDA Suite Overview

`src/eda.py` generates ten high-resolution PNG charts saved to `reports/eda_figures/`. Each chart answers a specific business or data-quality question.

Table 7.1: Complete EDA figure catalogue (10 figures)

Fig.	Title	Business Question Answered
01	Top Countries by Revenue	Which markets generate the most revenue outside the UK?
02	Monthly Revenue Trend	Is there seasonality? Are revenues growing?
03	Orders by Day of Week	Which weekdays have the highest order volume?
04	Quantity Distribution	How are order sizes distributed? Are there outliers?
05	Unit Price Distribution	What is the price range? Are there premium SKUs?
06	Top 10 Products by Revenue	Which SKUs drive the most revenue?
07	Customer Recency Distribution	How many customers are dormant?
08	RFM Pairplot	Are there visually separable clusters in the RFM space?
09	Elbow Curve (K vs WCSS)	What is the optimal number of clusters?
10	Cluster Profile Heatmap	How do the three segments differ across all features?

7.2 Key EDA Findings

- **Revenue concentration**: The UK accounts for $\approx$ 88% of total revenue. Among international markets, the Netherlands, EIRE, Germany and France are the top four.
- **Seasonality**: Monthly revenue peaks sharply in November 2011 ($\approx 3\times$ the January baseline), indicating strong Christmas gifting demand.
- **Recency skew**: Over 47% of customers last purchased more than 90 days before the end of the dataset window, signalling a large at-risk cohort.
- **RFM distributions**: All three RFM variables are strongly right-skewed. The pairplot reveals visually separable clusters after log-transformation, confirming that preprocessing is essential before clustering.

Chapter 8

Phase 3c – SQL Database Storage & Queries

8.1 Database Schema

`src/store_sql.py` creates a local SQLite database at `data/processed/retail.db` with two tables:

Table 8.1: SQLite database schema

Table	Source	Content
`transactions`	Phase 2 output	392,692 cleaned purchase lines with Revenue column
`customers`	Phase 3 output	4,338 customer feature profiles

8.2 Analytical SQL Queries

Five complex queries are executed automatically at pipeline runtime:

Table 8.2: Automated SQL analytical queries

ID	Label	SQL Logic
Q1	Top Spending Champions	`ORDER BY Monetary DESC LIMIT 10` – identifies highest lifetime-spend customers
Q2	Revenue by Country	`GROUP BY Country` with total revenue, order count, and unique customers
Q3	Monthly Revenue Trends	`strftime('%Y-%m')` grouping of transaction timestamps
Q4	Average RFM Metrics	Dataset-wide `AVG()` of all 7 features for benchmarking
Q5	Risky VIPs	Customers with Monetary > median **AND** CancellationRate > 0.20

The Q5 “Risky VIPs” query is strategically important: it surfaces customers who generate high revenue but also impose high return-processing costs, enabling targeted intervention before they become unprofitable.

Chapter 9

Phase 4 – Algorithm Selection & Justification

9.1 Candidate Algorithm Comparison

Four clustering paradigms were evaluated against our RFM feature space:

Table 9.1: Clustering algorithm comparison

Algorithm	Key Assumptions	Limitations for RFM	Suitability
K-Means	Spherical clusters of similar size; distance-based	Sensitive to outliers (mitigated by log1p + RobustScaler)	**High**
DBSCAN	Clusters separated by low-density regions	Merges overlapping customer density regions; hard to tune ϵ	Medium
GMM	Data follows Gaussian mixture distributions	Computationally expensive; overfits small datasets	Medium
Hierarchical	Distance hierarchy reflects true structure	$O(N^2 \log N)$ complexity; impractical for 4,338+ customers	Low

9.2 Evaluation Metrics

Three complementary metrics were used to select the optimal K and validate cluster quality:

Silhouette Coefficient (primary) $s(i) = \dfrac{b(i) - a(i)}{\max(a(i), b(i))}$, where $a(i)$ is the mean intra-cluster distance and $b(i)$ is the mean nearest-cluster distance. Range: $[-1, +1]$; higher is better.

Davies-Bouldin Index (secondary) Average similarity between each cluster and its most similar neighbour, computed from centroid distances and cluster scatter. Range: $[0, \infty)$; lower is better.

Calinski-Harabasz Score (tertiary) Ratio of between-cluster dispersion to within-cluster dispersion. Higher values indicate tighter, better-separated clusters.

9.3 Final Selection: K-Means

K-Means was selected because:

1. **Interpretability**: Centroids represent the "average customer" of each segment, directly translating to actionable marketing personas.
2. **Scalability**: Linear time complexity $O(I \cdot K \cdot N \cdot D)$ ensures fast execution even as transaction history grows.
3. **Preprocessing compatibility**: The log1p + RobustScaler preprocessing pipeline (Phase 5) fully mitigates K-Means' sensitivity to outliers and skewed distributions.

Chapter 10

Phase 5 – Machine Learning Clustering & Segment Profiling

10.1 Preprocessing Step 1: Log1p Transformation

All five right-skewed features are transformed using $\log(1 + x)$ before scaling:

Table 10.1: Skewness before and after log1p transformation

Feature	**Raw Skewness**	**Post-log1p Skewness**
Recency	1.83	0.61
Frequency	7.42	0.91
Monetary	9.17	0.74
AvgOrderValue	8.05	0.68
UniqueProducts	5.29	0.87
ActiveDays and CancellationRate are left untransformed (near-normal)		

Without this transformation, K-Means isolated a single cluster of top spenders and collapsed all remaining customers into one large undifferentiated group – a degenerate solution.

10.2 Preprocessing Step 2: RobustScaler

After log-transformation, `sklearn`'s `RobustScaler` normalises each feature using median and interquartile range (IQR), making the scaling robust to any remaining outliers:

$$X'_i = \frac{X_i - \text{median}(X)}{\text{IQR}(X)}$$

10.3 Hyperparameter Tuning

The primary hyperparameter for K-Means is the number of clusters K. A systematic grid search over $K \in \{2, 3, 4, 5, 6, 7, 8\}$ was conducted. Additional hyperparameters evaluated:

Table 10.2: K-Means hyperparameter tuning – before/after comparison

Parameter	Default	Final Value	Impact
K (clusters)	8 (max tested)	**3**	Silhouette +38% vs $K = 8$; most interpretable
`n_init`	10	10	Multiple restarts prevent poor local optima
`random_state`	None	42	Ensures exact reproducibility across runs
`max_iter`	300	300	Sufficient for convergence in all tested K

Table 10.3: Before/after: K=8 (default) vs K=3 (tuned)

Metric	K=8 (before)	K=3 (after)
Silhouette Score	0.2054	**0.3093**
Davies-Bouldin Index	1.9511	**1.2021**
Calinski-Harabasz Score	1,701.9	**2,341.8**
Cluster interpretability	Low (fragmented)	**High (clear personas)**

10.4 Optimal K Selection

K-Means was evaluated for $K \in \{2, 3, 4, 5, 6, 7, 8\}$:

Table 10.4: K selection metrics across candidate values

K	WCSS (Inertia)	Silhouette ↑	Davies-Bouldin ↓	Calinski-Harabasz ↑
2	11,204.3	0.2741	1.6120	1,847.3
lightblue **3**	**8,412.7**	**0.3093**	**1.2021**	**2,341.8**
4	7,108.5	0.2815	1.4302	2,198.5
5	6,341.2	0.2590	1.5887	2,051.2
6	5,892.4	0.2412	1.7001	1,923.4
7	5,543.1	0.2198	1.8230	1,803.7
8	5,287.9	0.2054	1.9511	1,701.9

$K = 3$ achieves the highest Silhouette Score, the lowest Davies-Bouldin Index, and the highest Calinski-Harabasz Score simultaneously – a clear and consistent optimum across all three independent metrics.

10.5 Segment Labelling Heuristic

Numerical cluster labels are mapped to business labels using a deterministic priority rule applied to cluster centroid values:

1. **VIP** ← cluster with the highest `Monetary` (highest total spend).
2. **Loyal** ← cluster with the lowest `Recency` among the remaining two (most recently active).
3. **At-Risk** ← the remaining cluster (highest recency = longest since last purchase).

This rule is implemented in `train_cluster_models.py::profile_clusters()` and is fully deterministic across re-runs.

10.6 Cluster Profiles

Table 10.5: Final cluster profiles – raw feature means

Segment	N	%	Recency	Freq.	Monetary	AOV	Products	Cancel
VIP	412	9.5	22.4d	87.3	£8,521	£97.6	312	0.082
Loyal	1,847	42.6	48.1d	31.6	£1,238	£39.2	98	0.061
At-Risk	2,079	47.9	187.3d	8.1	£298	£36.9	36	0.043

10.7 Feature Importance via Centroid Analysis

K-Means does not produce a direct feature importance score like decision trees. Instead, the relative importance of each feature is inferred from the **normalised centroid distances** – how much each feature's value differs between the three segment centroids.

Table 10.6: Feature importance ranking via centroid spread (coefficient of variation across 3 centroids)

Feature	Min centroid	Max centroid	Importance
Recency	22.4	187.3	High – primary separator between At-Risk and
Monetary	298	8,521	High – separates VIP from all others
Frequency	8.1	87.3	High – reinforces Monetary in VIP detection
ActiveDays	64.3	295.2	Medium – distinguishes long-term from casual
AvgOrderValue	36.9	97.6	Medium – separates VIP bulk buyers
UniqueProducts	36	312	Medium – breadth of catalogue engagement
CancellationRate	0.043	0.082	Lower – secondary risk signal within VIP clust

Key insight: Recency is the single most powerful separator, distinguishing At-Risk customers (187 days) from VIPs (22 days) with an 8× difference. Monetary is the primary VIP identifier, with a 29× spread between At-Risk and VIP centroids.

Chapter 11

Phase 6 – Model Card

11.1 Model Overview

Table 11.1: Model card – technical specification

Field	Value
Model name	RFM Customer Segmentation Pipeline
Model type	K-Means Clustering (unsupervised)
Version	1.0.0
Release date	17 June 2026
Authors	Hamza Amhidi & Mouaad Kaddouri
Serialisation	`models/best_clustering_pipeline.joblib`

11.2 Intended Use & Prohibited Applications

Table 11.2: Model use policy

Category	Details
Intended audience	Marketing teams, CRM managers, retail analysts
Supported decisions	Targeted email campaigns, loyalty reward allocation, win-back budgeting
Out of scope	Credit scoring, automated account suspension, individual price discrimination, fraud detection

11.3 Performance Summary

Table 11.3: Final model evaluation metrics

Metric	Interpretation	Value
Silhouette Score	Cluster cohesion and separation $\in [-1, +1]$	**0.3093**
Davies-Bouldin Index	Inter-cluster separation (lower = better)	**1.2021**
Calinski-Harabasz Score	Between/within dispersion ratio	**2,341.8**
Optimal K	Number of customer segments	**3**

11.4 Deployment Checklist

Table 11.4: Pre-deployment verification checklist

Requirement	Notes	Status
Monthly data refresh	Ingest updated transaction export	Required
7-feature parity check	Assert all columns present before inference	Automated
Scaler serialisation	RobustScaler saved in joblib bundle	Done
Segment label audit	Review VIP/Loyal/At-Risk thresholds quarterly	Manual
Data residency compliance	SQLite DB must remain on-premise or GDPR cloud	Required

11.5 Single-Command Execution

The entire pipeline is orchestrated by `run_pipeline.py`:

```
# Default: auto-download UCI dataset and run all phases
python run_pipeline.py

# Custom dataset (must match UCI schema)
python run_pipeline.py --input path/to/transactions.csv
```

Listing 11.1: Running the full pipeline

11.6 Output Artefacts

Table 11.5: output files

File	Description
`data/processed/customers_clustered.csv`	4,338 customers with Cluster and Segment labe
`models/best_clustering_pipeline.joblib`	Serialised log1p + RobustScaler + KMeans pip
`reports/eda_figures/09_elbow_curve.png`	WCSS + Silhouette vs K
`reports/eda_figures/10_cluster_profiles.png`	Normalised feature heatmap

Chapter 12

Results & Discussion

12.1 Final Segment Distribution

The pipeline produces three well-defined, business-interpretable segments from 4,338 customers:

Table 12.1: Segment summary and key characteristics

Segment	N	%	Profile	Recommended Strategy
VIP	412	9.5%	Purchased 22 days ago; 87 orders; £8,521 total	Premium loyalty programme, early product access, dedicated account manager
Loyal	1,847	42.6%	Purchased 48 days ago; 32 orders; £1,238 total	Upselling campaigns, volume discounts, mid-tier rewards
At-Risk	2,079	47.9%	Last purchase 187 days ago; 8 orders; £298 total	Win-back email sequences, personalised vouchers, re-engagement surveys

12.2 Discussion

Cluster quality: A Silhouette Score of 0.3093 is considered moderate-to-good for real-world customer data, where cluster boundaries are inherently soft. The Davies-Bouldin Index of 1.2021 (below the 1.5 practical threshold) and Calinski-Harabasz Score of 2,341.8 confirm that the three segments are meaningfully distinct.

Business insight: The At-Risk segment represents nearly half the customer base (47.9%), underscoring the scale of the churn risk. Even recovering 20% of At-Risk customers to the Loyal tier would represent a projected revenue uplift of approximately £62,000 annually based on average Loyal monetary values.

Preprocessing impact: The log1p transformation was the single most important engineering decision. Without it, K-Means collapsed all data into one large cluster with a degenerate Silhouette Score below 0.08. With it, the score jumped to 0.3093 – a 3.8× improvement – and segments became immediately interpretable.

Pipeline completeness: All 10 visualisation figures, 5 SQL analytical queries, the serialised model, and the full customer segment CSV are produced in a single `python run_pipeline.py` command.

Chapter 13

Conclusion

This project successfully delivered an end-to-end, reproducible customer segmentation pipeline that transforms 541,909 raw transaction records into three actionable customer segments: VIP (9.5%), Loyal (42.6%), and At-Risk (47.9%).

Key contributions:

- A seven-feature behavioural matrix extending classical RFM with AvgOrderValue, UniqueProducts, ActiveDays, and CancellationRate – with a corrected unique-invoice-level computation.
- A principled preprocessing pipeline (log1p $\rightarrow$ RobustScaler) that resolves the extreme skewness of retail transaction data.
- A rigorous K selection procedure validated by three independent metrics (Silhouette, Davies-Bouldin, Calinski-Harabasz).
- A fully serialised, CRM-ready pipeline with support for custom datasets.

Future work: (i) Monthly automated retraining as new transaction data becomes available. (ii) Adding geographic and temporal features. (iii) Extending to RFM scoring (ranked tiers) for finer-grained targeting.

Chapter 14

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